Deeksha Reddy Patlolla
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Software Developer with hands-on experience building and deploying **Java/Spring Boot microservices and REST APIs**. Delivered **production-grade enterprise applications** with a focus on CI/CD automation, cloud-native deployment, SQL performance, and distributed systems. Combines **strong Java, Spring Framework, and Azure cloud expertise** with proven experience in **JUnit/Mockito testing, Azure DevOps pipelines, and observability**.

EXPERIENCE

Data Engineer **March 2025 – Present**
University of Colorado, Denver | Denver, CO

- Designed and implemented **end-to-end ETL pipelines** using Python, SQL, and Apache Spark to ingest and transform data from multiple institutional source systems, supporting research analytics at scale.
- Developed and scheduled **automated data ingestion workflows** using Apache Airflow, reducing manual processing time by **50%** and ensuring consistent data freshness for downstream dashboards and reports.
- Designed normalized **relational data models** and integrated **Scikit-learn classification models** for student performance prediction, enabling proactive academic intervention and data-driven advising.
- Profiled and **tuned Apache Spark jobs and complex SQL queries,** identifying processing bottlenecks and reducing job runtimes, improving throughput across the analytics platform.
- Implemented automated **data validation checks and schema drift alerts within pipelines,** catching anomalies early and ensuring high data integrity for all downstream analytics consumers.
- Contributed to data warehouse schema design following star-schema and snowflake principles, improving **query performance** and simplifying data access patterns for reporting tools and analysts.
- Partnered with academic advisors and university administrators **to translate reporting requirements into technical data models** and pipeline specs, ensuring deliverables met real business needs.
- Authored **technical documentation for all ETL workflows**, data dictionaries, and ML integration specs, improving team onboarding speed and long-term platform maintainability.

Environment: Python, Java, SQL, Apache Spark, Apache Airflow, Scikit-learn, PostgreSQL, Azure SQL, Azure Monitor, Application Insights, ETL Pipelines, Data Warehousing

Software Engineer **Dec 2022 – May 2024**
Infor Pvt Ltd | Hyderabad, India

- Designed, developed, and maintained scalable **Java/Spring Boot microservices and RESTful service endpoints** for enterprise ERP applications using Spring MVC and Spring Framework (Core, MVC, Data, Security), improving system reliability and **reducing service downtime for 1,000+ business users.**
- Designed and implemented versioned **RESTful APIs for structured data workflows** integrated into production enterprise systems, with a strong focus on performance, security, and long-term maintainability.
- Wrote and optimized complex SQL queries, indexes, and stored procedures in Azure SQL/SQL Server and PostgreSQL, **reducing average query response time** and **improving data retrieval efficiency** across high-traffic service endpoints.
- Owned full lifecycle development of multiple product features , from requirements and design through implementation, testing, and production deployment , within Agile sprint cycles.
- Actively participated in peer code reviews, enforcing best practices around code structure, error handling, and test coverage, contributing to measurable reductions in post-release defects.
- Built and maintained **CI/CD pipelines using Azure DevOps and GitHub Actions**, automating build, test, package, and deployment stages; containerized Java services using Docker and supported deployments to Azure App Service and AKS.
- Integrated third-party APIs and internal microservices into the ERP platform; implemented authentication and authorization using Azure Active Directory (Entra ID) and OAuth2/OIDC patterns in Spring Boot applications, with Azure Key Vault for secrets management.
- **Wrote comprehensive unit and integration tests using JUnit and Mockito,** achieving high code coverage on critical service modules and significantly improving confidence in releases across the engineering team.

Environment: Java 8+, Spring Boot, Spring MVC, Spring Data, Spring Security, REST APIs, Azure (App Service, AKS, Entra ID, Key Vault), Azure DevOps, GitHub Actions, Docker, JUnit, Mockito, Azure SQL, PostgreSQL, SQL Server, Git, Maven, Agile/Scrum

TECHNICAL SKILLS

Languages	Java 8+/11/17, Python, SQL, C/C++, JavaScript, HTML/CSS
AI/ML & LLMs	PyTorch, TensorFlow/Keras, Scikit-learn, Hugging Face Transformers, LangChain, GPT-4o, YOLOv8, OpenCV
Data Engineering	Apache Spark, Airflow, Kafka, Redis, PostgreSQL, MySQL, MongoDB
Backend	Spring Boot, Spring MVC, Spring Data, Spring Security, Hibernate, JPA, FastAPI, Django, Node.js, REST APIs, GraphQL
Cloud & DevOps	Azure (App Service, AKS, SQL, Service Bus, Event Hubs, Monitor, Application Insights, Entra ID, Key Vault), Azure DevOps, GitHub Actions, Docker, Kubernetes, AWS, Git/GitHub, CI/CD, Linux/Unix

Databases	Oracle, My-SQL, SQL Server, Mongo DB, Cassandra

PROJECT EXPERIENCE

CivicLens – Policy Simulation Platform (Hackathon Winner) **April 2026**

- Built a **full-stack agentic AI platform** where policymakers upload policy documents; system pulls live data (US Census API, FRED API) and runs three parallel AI agents (Economist, Urban Planner, Equity Analyst) to simulate real-world impact.
- Engineered **RAG pipeline** (ChromaDB, sentence-transformers) and LangGraph orchestration; rendered outcomes on an interactive 3D Mapbox map (Next.js, TypeScript, Tailwind CSS); reduced response latency from **~45s to ~15s** via parallel agent execution.

Heterogeneity-Aware Multimodal Learning for Early Disease Detection **Dec 2025**

- Designed a **transformer fusion model** integrating imaging, genomic, and clinical data using PyTorch, improving ROC-AUC by 11% over early-fusion baselines.
- Built an **end-to-end training pipeline** with gated attention, cross-validation, and SHAP-based interpretability for modality-level contribution analysis.

Resume AI Intelligence Platform (RAG + Multi-Agent LLM) **Nov 2025**

- Designed and deployed **a multi-agent LLM system** using LangChain + GPT-4 with a RAG pipeline powered by pgvector/Supabase for semantic resume parsing and job matching.
- Engineered a **production-ready FastAPI backend** with memory-aware context handling, reducing token usage by 28%.

Autonomous Cybersecurity Defender **Sep 2025**

- Built a **real-time intrusion detection system** combining rule-based filters with GPT-4o classification, trained on CICIDS2017 dataset achieving **94%** detection accuracy on intrusion detection tasks.
- Engineered a FastAPI backend with **2.5s response latency** and validated endpoints using Thunder Client.

EDUCATION

Master of Science in Computer Science – GPA: 3.8 **Aug 2024 – May 2026**
University of Colorado, Denver | Denver, CO

Bachelor of Technology in Computer Science – GPA: 3.6 **Aug 2020 – May 2024**
Malla Reddy College of Engineering and Technology | Hyderabad, India

PUBLICATIONS

Unmasking Digital Deception: Detecting Deepfakes through Deep Learning
IRJMETS, 2024

- Developed a **hybrid CNN-RNN architecture (ResNeXt + LSTM)** for deepfake detection, capturing both spatial and temporal inconsistencies in video data.
- Built a **multi-dataset training pipeline** (DFDC, FaceForensics++, Celeb-DF) to improve generalization and real-world detection performance.